Higher Structures in Rome

Schedule & Abstracts

Thursday, 17 September		Friday, 18 September	
10:25–10:30	Welcome!	09:30–10:30	Horel
10:30–11:30	Muro	10:30–11:00	Coffee Break
11:30–12:00	Coffee Break	11:00–12:00	Cirici
12:00–13:00	Pratali	12:10–13:10	Boavida
13:00–15:00	Lunch Break	13:10–15:00	Lunch Break
15:00–16:00	Malin	15:00–16:00	Stiénon
16:00–16:30	Coffee Break	16:00–16:30	Coffee Break
16:30–17:30	Krähmer	16:30–17:30	Rossi
17:40–18:40	Fiorenza	17:40–18:40	Vallette
		20:30–23:59	Conference Dinner

Pedro Boavida (Lisbon)

Cohomological obstructions to isotopy

In the so-called metastable range, Haefliger and Becker showed that the problem of determining whether two smooth embeddings are isotopic is controlled by a single cohomology class. I will explain, with examples, how embedding calculus generalises this by providing not one but a finite list of obstruction classes, which form a complete obstruction to isotopy in codimension at least three. These obstruction classes live in cohomology groups of configuration spaces with coefficients in certain parameterised spectra whose fibres are closely related to the spectral Lie operad. This is joint work with Geoffroy Horel and Michael Weiss.

Joana Cirici (Barcelona)

Configuration spaces of algebraic varieties

I will use the theory of weights in étale cohomology to give a simple and conceptual proof of a theorem of Kriz stating that a rational model for the ordered configuration space of a smooth complex projective variety is given by the second page of the Leray spectral sequence for the rational constant sheaf relative to the obvious inclusion, along with its only non-trivial differential. Our proof builds on Totaro's study of this spectral sequence, combined with a basic observation related to the formality of filtered dg-algebras. An advantage of this proof is that it allows for a generalization to study the p-adic homotopy type of configuration spaces for certain algebraic varieties defined over finite fields. This is joint work with Geoffroy Horel.

Domenico Fiorenza (Roma Sapienza)

Batalin-Vilkovisky algebras from Ševera bicomplexes

A Ševera bicomplex is a bicomplex (M, d_0, d_1) whose differentials d_i are Grothendieck differential operators of order $\leq i$ with respect to a given A-module structure on M, where A is a graded commutative algebra. One also requires that (M, d_0) admits a contracting homotopy K that is a differential operator of the same order as d_0. Under suitable assumptions, the differential d_2 of the second page of the spectral sequence associated with (M, d_0, d_1) is a Batalin-Vilkovisky operator on E_2, so that when E_2 is a free rank one A-module, the graded commutative algebra structure of A is enhanced to a Gerstenhaber algebra structure, and the choice of a d_2-closed basis element for E_2 further enhances this to a Batalin-Vilkovisky algebra structure. The prototypical example of this construction is Ševera's description ofthe Batalin-Vilkovisky algebra structure on the algebra of smooth functions of an a odd symplectic manifold. Joint work with Eugenia Boffo.

Geoffroy Horel (Sorbonne Paris Nord)

Equivariant formality of little disks operads

I will explain a proof of the following theorem: the little n-disks operad is equivariantly formal with respect to the action of the group $SO(n)$. As a consequence, the framed little disks operad is also formal. Part of the talk will be devoted to give a precise formulation of the theorem which is a little bit more subtle than one may think. Then I will explain the proof which relies on an action by the Grothendieck-Teichmüller group and a quite general formality machinery. This is joint work with Pedro Boavida de Brito and Joana Cirici.

Ulrich Krähmer (Dresden)

Clones from Comonoids

The fact that the cocommutative comonoids in a symmetric monoidal category form the best possible approximation by a cartesian category is revisited when the original category is only braided monoidal. This leads to the question when the endomorphism operad of a comonoid is a clone (a Lawvere theory). By giving an explicit example, we prove that this does not imply that the comonoid is cocommutative (joint with Myriam Mahaman).

Connor Malin (MPI Bonn)

Formal geometric Lie algebras: a homotopical perspective on complete Lie algebras

Mandell's Theorem asserts that the category of finite-type nilpotent and p-complete spaces fully faithfully embeds into the category of E_∞-algebras over an algebraically closed field of characteristic p. There are major obstructions to proving Mandell's theorem using L_∞-algebras since Koszul duality is very degenerate when restricted to the E_∞-algebras associated to spaces in characteristic not equal to 0. Roca i Lucio has given an interesting point-set solution to this problem by transferring the model structure on E_∞-algebras to a certain category of complete L_∞-algebras. In this talk, I will give a homotopical solution by completing the ∞-category of spectral Lie algebras to a certain category of "Formal Geometric Lie algebras". We will give a construction of a space's "tangent formal geometric Lie algebra" over the sphere spectrum and demonstrate that it agrees with a completion of the cotangent complex of the space's spherical chains.

Fernando Muro (Seville)

Kadeishvili's formality theorem and universal Massey products

In the 1980s, Kadeishvili proved two by now classical theorems. The first one says that a differential graded algebra is, up to quasi-isomorphism, determined by the transferred minimal A-infinity structure carried by its cohomology. The second one establishes that, under suitable Hochschild-cohomological vanishing conditions, the minimal A-infinity structure is necessarily trivial, and hence unique: this is Kadeishvili's intrinsic formality theorem. In this talk, I will present a generalization of Kadeishvili's intrinsic formality theorem, obtained jointly with Gustavo Jasso (Cologne). This generalization is suitable for non-formal differential graded algebras. It brings into play, as part of the input, the first obstruction to formality, the universal Massey product, which lives in Hochschild cohomology. This Hochschild cohomology class has previously been considered by Baues-Dreckmann, Benson-Krause-Schwede, and Kaledin, among others. In addition, we show how to leverage the full power of the Gerstenhaber algebra structure on Hochschild cohomology to classify differential graded algebras for which the universal Massey product is unit-like. Moreover, when the cohomology algebra is shifted self-dual as a bimodule, we show that these differential graded algebras satisfy the Calabi-Yau property if and only if the Batalin-Vilkovisky operator associated with the duality kills the universal Massey product. These conditions arise naturally in representation theory. The results were developed in this setting in order to prove the derived Auslander-Iyama correspondence, a joint work with Jasso which classifies differential graded algebras whose regular representation has a cluster-tilting property. The Calabi-Yau refinement

uses the corresponding Batalin–Vilkovisky structure and leads, in particular, to examples of Calabi–Yau triangulated categories which cannot be lifted to the differential graded level.

Francesca Pratali (Utrecht)

Good completion for Koszul operads

Koszul duality provides a powerful way to relate algebras over an operad to coalgebras over its bar construction. In the setting of stable ∞-categories, this duality is closely tied to two natural notions of completion for operadic algebras: nilpotent completion, arising from the tower of truncations of an operad, and homological completion, defined with respect to the cotangent fiber functor. Although these completions agree in important examples, they need not coincide in general.

A central role in understanding this comparison is played by the notion of good completion for an operad O, introduced by Heuts. Roughly, this requires every trivial O-algebra to be nilcomplete. When this condition holds, nilpotent and homological completion agree, and the resulting Koszul duality formalism acquires particularly strong properties.

While good completion is known to hold for some classes of operads, the scope of this property is still largely unexplored, and it is still unclear how broadly it holds for operads of interest. In this talk, we prove that every binary quadratic operad over a field of characteristic zero which is Koszul in the sense of Ginzburg–Kapranov or Loday–Vallette has good completion. As a consequence, we deduce that the little n-disks operads have good completion over the rational numbers. This is joint work with M. Barata, M. Carli, and G. Heuts.

Tommaso Rossi (Bologna)

Koszul duality: old and new results

Koszul duality can be seen as a bridge connecting two seemingly unrelated objects, and has various manifestations in algebra and topology. An important example of this is the duality between commutative and Lie algebras, which is crucial to connect Sullivan's and Quillen's approach to rational homotopy theory. In this talk we will review the basics of the theory and its main applications. Then we will focus on the duality between the hypercommutative and gravity operad, two operads related to the moduli spaces of genus zero curves. At the homological level, this duality was proved by Getzler. At the chain level, this is part of a joint work with Paolo Salvatore.

Mathieu Stiénon (Penn State)

A Duflo–Kontsevich-type theorem for differential graded manifolds

Differential graded (dg) manifolds provide a powerful framework unifying diverse algebraic and geometric structures including homotopy Lie algebras, foliations, complex manifolds, and spaces with singularities. For any dg algebra, the pair consisting of its Hochschild cohomology and homology carries a rich algebraic structure reminiscent of the classical Cartan calculus and known as the Tamarkin–Tsygan calculus. I will present a Duflo–Kontsevich-type theorem in the setup of the Tamarkin–Tsygan calculus associated with the dg algebra of functions on a dg manifold. Specializing this general result to two important classes of dg manifolds, one recovers both Duflo's classical theorem on the center of the universal enveloping algebra of a Lie algebra and Kontsevich's theorem on the Hochschild cohomology of complex manifolds. This is joint work with Hsuan-Yi Liao and Ping Xu.

Bruno Vallette (Sorbonne Paris Nord)

Explicit formulas for the noncommutative Cartan calculus

The commutative/classical Cartan calculus amounts to a Lie type double action of vector fields of a smooth manifold on its de Rham complex of differential forms. Its noncommutative analogue is expected to take the form of a homotopy Lie type double action of the Hochschild cochain complex of a (homotopy) associative algebra on its the Hochschild chain complex compatible with Connes' boundary map. Such a structure

implies a noncommutative chain-level version of the Gauss-Manin connection. In this talk, I will explain how the operadic calculus allows one to solve this problem and I will provide fully explicit formulas in terms of an operad introduced by Kontsevich-Soibelman.